

## Convolution

# 82Bh

$$h[n] = 0.5^n (u[n] - u[n-3]) = \{1, 0.5, 0.25\}$$

$$x[n] = \{2, 1, 0, 5, -1\}$$

Convolution:

$$y = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$$

$$= \sum_{k=0}^{\infty} x[k] h[n-k]$$

$$= x[0] h[n] + x[1] h[n-1] + x[2] h[n-2] + x[3] h[n-3] + x[4] h[n-4]$$

$$= 2h[n] + h[n-1] + 5h[n-3] - h[n-4]$$

$$y[0] = 2h[0] = 2$$

$$y[1] = 2h[1] + h[0] = 2 \times 0.5 + 1 = 2$$

$$y[2] = 2h[2] + h[1] = 2 \times 0.25 + 0.5 = 1$$

$$y[3] = 2h[3] + h[2] + 5h[0] = 0.25 + 5 \times 1 = 5.25$$

$$y[4] = 5h[1] - h[0] = 5 \times 0.5 - 1 = 1.5$$

$$y[5] = 5h[2] - h[1] = 5 \times 0.25 - 0.5 = 0.75$$

$$y[6] = -h[2] = -0.25$$

# 82 Ba

$$h[n] = \left(\frac{1}{2}\right)^n u[n]$$

$$x[n] = 5e^{j\frac{\pi}{3}n}$$

convolution:

$$y = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$$

$$= \sum_{k=0}^n x[k] h[n-k]$$

$$= \sum_{k=0}^n 5e^{j\frac{\pi}{3}k} \cdot \left(\frac{1}{2}\right)^{n-k}$$

$$= 5 \times \left(\frac{1}{2}\right)^n \sum_{k=0}^n (e^{j\frac{\pi}{3}} \cdot 2)^k$$

$$= \frac{5}{2^n} \times \frac{(2e^{j\frac{\pi}{3}})^{n+1} - 1}{2e^{j\frac{\pi}{3}} - 1}$$

$$= \frac{5}{2^n} \left( \sum_{k=1}^n (2e^{j\frac{\pi}{3}})^k + 1 \right) \quad \checkmark \quad a^{\frac{r^n-1}{r-1}}$$

$$= \frac{5}{2^n} \times \left[ \frac{(2e^{j\frac{\pi}{3}})^n \times (2e^{j\frac{\pi}{3}})^{n-1} + 1}{2e^{j\frac{\pi}{3}} - 1} \right]$$

$$= \frac{5}{2^n} \times \left[ \frac{(2e^{j\frac{\pi}{3}})^{n+1} - (2e^{j\frac{\pi}{3}})^0 + 1}{2e^{j\frac{\pi}{3}} - 1} \right]$$

$$= \frac{5}{2^n} \times \left[ \frac{(2e^{j\frac{\pi}{3}})^{n+1} - (2e^{j\frac{\pi}{3}}) + 2e^{j\frac{\pi}{3}} - 1}{2e^{j\frac{\pi}{3}} - 1} \right]$$

$$= \frac{5}{2^n} \times \frac{(2e^{j\frac{\pi}{3}})^{n+1} - 1}{2e^{j\frac{\pi}{3}} - 1}$$

# 8L Bh

$$h[n] = 2\delta[n+1] + 2\delta[n-1] = \{2, 0, 2\}$$

$$x[n] = \delta[n] + 2\delta[n-1] - \delta[n-3] = \{1, 2, 0, -1\}$$

convolution:

$$y = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$$

$$= \sum_{k=-\infty}^{\infty} x[k] h[n-k]$$

$$= x[0]h[n] + x[1]h[n-1] + x[2]h[n-2] + x[3]h[n-3]$$

$$= h[n] + 2h[n-1] - h[n-3]$$

$$y[0] = h[0] + 2h[-1] = 4$$

$$y[1] = h[1] + 2h[0] = 2$$

$$y[2] = h[2] + 2h[1] - h[-1] = 2$$

$$y[3] = 2h[2] - h[0] = 0$$

$$y[4] = -h[2] = -2$$

# 8/18a

$$h[n] = \left(\frac{1}{3}\right)^n u(n-3)$$

$$x[n] = \left(\frac{1}{6}\right)^{n-6} u(n)$$

convolution:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$$

$$= \sum_{k=0}^{\infty} \left(\frac{1}{6}\right)^{k-6} h[n-k]$$

$$= 6^6 \sum_{k=0}^{\infty} \left(\frac{1}{6}\right)^k h[n-k]$$

$$= 6^6 \left[ \frac{1}{6^0} \times 0 + \frac{1}{6^1} \times 0 + \frac{1}{6^2} \times 0 + \frac{1}{6^3} \times h[n-3] + \frac{1}{6^4} \times h[n-4] \right]$$

$$= 6^6 \sum_{k=3}^{\infty} \frac{1}{6^k} \left(\frac{1}{3}\right)^{n-k}$$

$$= \frac{6^6}{3^n} \sum_{k=3}^{\infty} \frac{1}{6^k} \times 3^k$$

$$= \frac{6^6}{3^n} \sum_{k=3}^{\infty} \frac{1}{2^k}, \text{ let } k-3=k_1, \quad k=k_1+3$$

$$= \frac{6^6}{3^n} \sum_{k_1=0}^{\infty} \left(\frac{1}{2}\right)^{k_1+3}$$

$$= \frac{6^6}{3^n} \times \frac{1}{2^3} \times \sum_{k_1=0}^{\infty} \left(\frac{1}{2}\right)^{k_1}$$

$$= \frac{6^6}{3^n} \times \frac{1}{2^3} \times 2$$

$$= \frac{3^6 \times 2^6}{3^n \times 2^3} \times 2 = 3^{6-n} \times 2^9 = 16 \times 3^{6-n}$$

# 80 Ba  
→ 82 Ba

# 79 Rh  
→ 81 Rh

# 78 Bh

$$h[n] = u[n] - u[n-4] = \{4, 4, 4, 1\}$$

$$x[n] = \left(\frac{1}{2}\right)^n u[n] = \left(\frac{1}{2}\right)^n, n \geq 0$$

Here,

$$y[n] = \sum_{k=-\infty}^{\infty} h[k] x[n-k]$$

$$= \sum_{k=0}^{\infty} h[k] x[n-k]$$

$$= h[0]x[n] + h[1]x[n-1] + h[2]x[n-2] + h[3]x[n-3]$$

$$= x[n] + x[n-1] + x[n-2] + x[n-3]$$

$$y[0] = x[0] = 1$$

$$y[1] = x[1] + x[0] = 1.5$$

$$y[2] = x[2] + x[1] + x[0] = 1.75$$

$$y[3] = x[3] + x[2] + x[1] + x[0] = 1.875$$

$$y[4] = x[4] + x[3] + x[2] + x[1]$$

$$\therefore y = \begin{cases} 0 & n < 0 \\ 1 & n = 0 \\ 1.5 & n = 1 \\ 1.75 & n = 2 \\ 1.875 & n = 3 \\ \vdots & n \geq 4 \end{cases}$$

other papers are repeated/  
same concept.